

LC 1377: Frog Position After T Seconds

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Problem Constraints & Initial Thoughts

Morgan Stanley

Arcesium

1 to n, undirected graph, frog can jump from one vertex to another unvisited vertex.

(Root Node)
/ | \
P=(0.333) P=(0.333) P=(0.333)

equal property

We are given t jumps. If no node connected and jumps still left, frog will simply jump in place.

We need to tell that after t seconds the frog is on vertex target.

Approach & State Formulation

We can have something like:

- $dp[node][time]$ → to store the probability of being at node at time t

OR

In BFS:

queue state can be: curr node, curr time, curr prob

- Initially → (1, 0, 1)
- and then → child node (2, 1, $1 \times 1/\text{child}$)
↑ (0+1)

If DP $\rightarrow$ `dp[child][time+1] += dp[parent][time] / children`

C++ Implementation

Notes:

- basic aadya to bfs krne ka hi he
- queue state me node, currttime, currprobab

```
class Solution {
public:
    double frogPosition(int n, vector<vector<int>>& edges, int t, int target)
    {
        //basic aadya to bfs krne ka hi he
        //queue state me node,currttime,currprobab
        vector<vector<int>> adj(n+1);
        vector<int> seen(n+1,0);

        for(auto &it:edges){
            int u=it[0];
            int v=it[1];

            adj[u].push_back(v);
            adj[v].push_back(u);
        }

        queue<pair<int,pair<int,double>>> q; //queue me basically
                                           //{currnode,{currt,currprobab}}

        q.push({1,{0,1.0}});
        seen[1]=1;

        while(!q.empty()){
            int currn=q.front().first;
            int currt=q.front().second.first;
            double currp=q.front().second.second;

            q.pop();

            if(currt==t && currn==target){
                return currp;
            }

            int child=0;
```

```

        for(auto &it:adj[currn]){
            if(!seen[it]){ //cause parent cant be child
                child++;
            }
        }

        if(child==0){
            //wahi pe jump krte rhega
            if(currt<t)
                q.push({currn,{currt+1,currp}});
        }
        else{
            //nahi to child me kudega
            for(auto &it:adj[currn]){
                if(!seen[it]){
                    seen[it]=1;
                    q.push({it,{currt+1,(double)(currp/(child*1.0))}});
                }
            }
        }
    }

    return 0.0;
}

};

```